

Interacting Multipoles of Rank 2 and 2:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \end{array} \right] \quad (1)$$

Simplify deltas

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{array} \right] \quad (2a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{array} \right] \quad (2b)$$

Simplify R

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Theta_{zz} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{array} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^4 \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^4 \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^4 \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^4 \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{array} \right] \quad (3b)$$

Simplify Multipoles

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \quad (4a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \quad (4b)$$

Exploit Traceless Conditions

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \end{bmatrix} \quad (5a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \end{bmatrix} \quad (5b)$$

Reduce Indices

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \end{bmatrix} \quad (6a)$$

$$\xrightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \end{bmatrix} \quad (6b)$$

Add up

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[+45 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \right] \quad (7)$$

Combining everything

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = \left[-3 \cdot R_z^{-5} \cdot \Theta_{zz} \cdot \Theta_{zz} \right] \quad (8)$$